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Psychological aspects of tinnitus and the application of cognitive–behavioral therapy

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Abstract

This article presents an overview of tinnitus (ringing or buzzing in the ears), its psychological effects, and the application of cognitive–behavioral therapy (CBT) for its treatment. Several studies have confirmed an association between psychological factors, such as anxiety and depression, and severe tinnitus and preliminary reports suggest that a proportion of tinnitus patients suffer from mental illness. Assessment strategies used in CBT for tinnitus include structured interviews, daily diary ratings, and validated self-report questionnaires. The treatment approach described in this article includes applied relaxation, imagery and distraction techniques, advice regarding environmental sounds, management of sleep, cognitive restructuring of thoughts and beliefs associated with tinnitus, and relapse prevention. The literature pertinent to CBT approaches to treating tinnitus is reviewed, and it is concluded that CBT shows promise as a treatment of tinnitus-related distress. Future research directions are discussed.

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1. Introduction

Tinnitus is an “auditory phantom perception” defined as the experience of sound in the absence of any appropriate external stimulus (Tyler, 2000). The sounds may be very loud and

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bizarre, and the most common ones are like a musical tone, or a rushing sound like escaping steam, or air. Tinnitus is known to occur as a concomitant of almost all the dysfunctions that involve the human auditory system. Hence, damage to the middle ear, the cochlea, the VIII cranial nerve (audiovestibular), and pathways in the brain from cochlear nucleus to primary auditory cortex are likely candidates for explaining tinnitus (McFadden, 1982; Tyler & Babin, 1993). However, the most common cause of tinnitus is hearing loss and, not surprisingly, a majority of people with tinnitus have impaired hearing (Tyler, 2000).

While tinnitus is typically a temporary sensation, it may develop into a chronic condition that is resistant to medical or surgical treatment. Prevalence figures suggest that at least 10% of the general population has suffered from tinnitus, although only 1–3% has it in its severe form (Davies & Rafie, 2000). Severe tinnitus causes disruption of everyday activities, mood, sleep patterns; is often chronic; and is more common in adults and in the elderly than in children (Davies & Rafie, 2000). While gender differences have been observed—for example, women have been reported to have more complex tinnitus sounds (Dineen, Doyle, & Bench, 1997; Meikle & Greist, 1989)—there are no clear indications that tinnitus should be experienced differently by men and women.

Severe tinnitus is often distinguished by subjective reports of intrusiveness and irritation because of the sounds (McKenna & Andersson, 1998). However, it has not been possible to determine the handicap caused by tinnitus using the characteristics of the tinnitus sound itself (e.g., loudness, character), and tinnitus has been notoriously difficult to measure objectively (Tyler, 2000). Psychological complaints have been associated with severe tinnitus, and the role of psychological factors was observed early (Fowler, 1948). For example, insomnia is a significant problem for many tinnitus patients (McKenna, 2000). In a clinical sample of 216 patients, Andersson, Lyttkens, and Larsen (1999) found that 153 (71%) reported sleep problems. This is a higher percentage than reported by other researchers, who usually find between 25% and 50% suffering from sleep problems (Sanchez & Stephens 1997; Tyler & Baker 1983). Tinnitus patients also often report difficulties with concentration, as evidenced by reading performance (Sanchez & Stephens, 1997; Tyler & Baker, 1983). Until recently, there have been few attempts to measure tinnitus patients' performance on tests of cognitive functioning, but preliminary results corroborate the self-report findings (Andersson, Eriksson, Lundh, & Lyttkens, 2000; McKenna & Hallam, 1999; McKenna, Hallam, & Shurlock, 1996).

Most clinicians agree that tinnitus needs to be managed within a multidisciplinary setting. Recently, Dobie (1999) reviewed several controlled treatment trials and concluded, “Non-specific support and counseling are probably helpful” (p. 1202), and that medical treatments seldom cure tinnitus. Given that many of the problems experienced by tinnitus sufferers (e.g., depression, anxiety, insomnia) have been successfully targeted with cognitive-behavioral therapy (CBT) clinicians in well-controlled studies (Hawton, Salkovskis, Kirk, & Clark, 1989), CBT is likely to be of benefit in tinnitus management. The effect of CBT on coping with tinnitus has been reviewed previously (Andersson & Lyttkens, 1999; Henry & Wilson, 2000). The aim of this article is to present a broader and updated overview of the psychological effects of tinnitus and how CBT can be applied in the treatment of tinnitus.

2. Psychological states and traits associated with tinnitus

Several studies have been devoted to the psychological problems associated with tinnitus, focusing on sleep problems, concentration difficulties, anxiety, depression (and suicide), stress, and personality characteristics of tinnitus patients (Erlandsson, 2000). Alternative frameworks have been applied in this research, including personality psychology, interpersonal attachment theory, neuropsychology, and psychiatric nomenclature according to the DSM (American Psychiatric Association, 1994). Effects of tinnitus on psychological functioning, and vice versa, have been assessed via self-report questionnaires, performance tests of cognitive functioning, and structured interviews. This section focuses on studies using self-report and structured clinical interviews.

While preexisting psychological characteristics most likely affect the way a person reacts to tinnitus (Stouffer & Tyler, 1992), there is little evidence that psychiatric disturbance is a direct cause of tinnitus. Andersson and McKenna (1998) applied a diathesis–stress model (see also Schulman, 1995), suggesting that a vulnerable person might develop tinnitus distress following a comparatively low degree of tinnitus. On the other hand, a more stress-tolerant person might bear higher degrees of tinnitus before seeking help. In line with this, Attias et al. (1995) found that help-seekers displayed more psychiatric symptoms than non-help-seekers. Nonetheless, the tinnitus patients who did not seek help had more psychological problems than a control group without tinnitus. Curiously, the perceived loudness of tinnitus was significantly lower in the help-seeking group. Differences between tinnitus “complainers” and “noncomplainers” have previously been found on psychological measures (Hallberg & Erlandsson, 1993; Kirsch, Blanchard, & Parnes, 1989).

The personality of the tinnitus patient has also been the subject of some interest (e.g., Gerber, Nehemkis, Charter, & Jones, 1985). For example, Collet, Moussu, Disant, Ahami, and Morgon (1990) used the Minnesota Multiphasic Personality Inventory (Butcher, Dahlstrom, Graham, Tellegen, & Kaemmer, 1989) and found normal scores overall, with the exception of elevated depression scores in men. Dispositional optimism, as assessed by the Life Orientation Scale (LOT) (Scheier & Carver, 1985) was found to be negatively related with tinnitus complaints in a small study (Andersson, 1996a). This finding suggests that individual differences in coping ability are related to dispositional personality characteristics such as optimism. This relationship has also been observed for other conditions, such as coronary heart disease (Andersson, 1996b).

A relationship between tinnitus and psychological distress has now been observed in several studies (e.g., Folmer, Greist, Meikle, & Martin, 1999; Halford & Anderson, 1991; McKenna, Hallam, & Hinchcliffe, 1991). Most of the available evidence has been based on questionnaire data, but there are a few small studies that have employed structured psychiatric interviews adopting DSM criteria (American Psychiatric Association, 1994). Harrop-Griffiths, Katon, Dobie, Sakai, and Russo (1987) interviewed 21 consecutive patients and found a lifetime prevalence of major depression in 62% and current depression in 48%. Simpson, Nedzelski, Barber, and Thomas (1988) found that 63% of their tinnitus sufferers ($N=24$) could be classified as suffering from a DSM-III-R defined psychiatric disturbance and, of these, 46% had a diagnosable mood disorder. In a sample of 40 consecutive tinnitus patients,

Sullivan et al. (1988) found a lifetime prevalence for depression of 78%. Importantly, at the time of the interview, 60% of the patients had signs of depression. Hiller and Goebel (1992) found that 26 out of 27 tinnitus patients fulfilled the DSM-III-R criteria for at least one lifetime diagnosis. Russo, Katon, Sullivan, Clark, and Buchwald (1994) used the DSM-IV to estimate the prevalence of psychiatric disorder among 88 tinnitus patients. No specific figures were provided for separate diagnoses, but it was reported that the average number of current and past diagnoses were 0.7 and 0.6, respectively. Most recently, Zöger, Holgers, and Svedlund (2001) studied a group of 82 tinnitus patients using DSM-IV diagnostic criteria. Results showed that 62% of the patients had signs of lifetime depressive illness and that 39% had an ongoing depression. Other diagnoses were rarer, with the exception of anxiety disorders (45%).

Despite its association with depression, suicide attributable to tinnitus appears to be uncommon. Most suicide cases reported in association with tinnitus have had comorbid psychiatric disturbances (Lewis, Stephens, & Huws, 1992; Lewis, Stephens, & McKenna, 1994). Lewis and Stephens (1996) have reported that suicide attempters rarely indicate that tinnitus was a problem for them; in fact, tinnitus was less common than in the general population, with only 1.6% reporting tinnitus (compared to the prevalence of 10% commonly found in epidemiological surveys). However, according to Johnston and Walker (1996), older males may be at risk for committing suicide in association with tinnitus. Suicide should always be taken into account as thoughts about suicide do occur in association with severe tinnitus (Stephens, Erlandsson, Sanchez, & Huws, 1992).

A few studies have investigated anxiety in tinnitus sufferers, finding slightly elevated levels on questionnaire measures (Briner, Risey, Guth, & Noris, 1990; Halford & Anderson, 1991; Stephens & Hallam, 1985). Somatic attention and related anxiety have recently been suggested as important factors in determining whether tinnitus is perceived as annoying (Heath, 1994; Hiller, Janca, & Burke, 1997; Newman, Wharton, & Jacobson, 1997). Tinnitus often becomes a problem when it is first attended to, and many people manage to distract themselves from the sounds. Newman et al. (1997) found an association between somatic attention and distress caused by tinnitus.

Recently, Andersson and Vretblad (2000) studied the construct of anxiety sensitivity in tinnitus sufferers. Anxiety sensitivity is defined as “fear of anxiety-related symptoms that result from belief that these symptoms have harmful somatic, social, or psychological consequences” (Taylor, Koch, McNally, & Crockett, 1992, p. 245). Anxiety sensitivity is usually measured with an instrument called the Anxiety Sensitivity Index (ASI; Reiss, Peterson, Gursky, & McNally, 1986). In a study with 146 tinnitus patients with moderate to severe tinnitus, anxiety sensitivity correlated significantly with tinnitus distress ($r=.60$), and the association was significantly stronger in female participants who also scored higher on the ASI (Andersson & Vretblad, 2000). In addition, in a recent controlled treatment study, scores on the ASI decreased significantly following treatment (Andersson, Strömberg, Ström, & Lyttkens, *In Press*). The treatment consisted of Internet-based self-help, supplemented with e-mail. In the treatment group, mean scores on the ASI decreased significantly from 20.6 pretreatment to 16.2 at posttreatment. At 1-year follow-up, results were maintained with a mean score of 16.8. Unfortunately, this score was not statistically significant from the

pretreatment scores. The corresponding ASI scores in the control group were 19.1 and 17.8 at pre- and posttreatment, respectively, and at follow-up, after having received treatment, it was 15.3.

A rather neglected topic is how tinnitus affects the family and primary relations surrounding the individual with tinnitus. Recently, [Granqvist, Lanto, Ortiz, and Andersson \(2001\)](#) used a tinnitus-adapted version of the Family Support Scale of the West Haven–Yale Multidimensional Pain Inventory ([Kerns, Turk, & Rudy, 1985](#)) with the purpose of measuring perceptions of family members responses to the individual with tinnitus (see also [Sullivan, Katon, Russo, Dobie, & Sakai, 1994](#)). In addition, they used a self-report measure of adult attachment—the Adult Attachment Scale ([Collins & Read, 1990](#)). The study included 102 tinnitus patients who had previously visited the audiology department as outpatients. Results using multiple regression analyses showed avoidant attachment to be related to both tinnitus-related problems and perceived punitive family responses, ambivalent attachment to be linked to the latter, and secure attachment to be unrelated to both ([Granqvist et al., 2001](#)).

In sum, the prevalence of psychiatric problems among tinnitus patients has not been researched in depth. Several studies confirm a link between tinnitus and emotional disturbances in some patients. There is a lack of studies that address the association between psychiatric diagnoses and tinnitus in the general population and a dearth of long-term follow-up studies of psychological morbidity in association with tinnitus. As sampling bias may inflate the association, an area requiring urgent attention is the investigation of the link between psychiatric disturbance and tinnitus in random samples.

3. Psychological assessment of tinnitus patients

Assessment devices used in the psychological management of tinnitus can be divided into those that are tinnitus-specific and those that are global (e.g., those aimed at assessing psychological well-being and cognitive functioning). In this context, it is important to be aware of the special needs and circumstances surrounding the tinnitus patient, which may affect the ability to complete assessment. The most obvious example is hearing impairment, a common comorbid problem. However, the assessor also needs to be vigilant toward other less conspicuous presenting problems, such as fatigue, lack of motivation, litigation status, and the like. All of these hold potential to affect the assessment results.

The psychological dimensions of tinnitus complaints are usually measured with self-report instruments specifically developed for use with tinnitus patients ([Noble, 2000; Tyler, 1993](#)). One example is the Tinnitus Reaction Questionnaire (TRQ) developed by [Wilson, Henry, Bowen, and Haralambous \(1991\)](#). The TRQ has been used in many research studies and has been translated into several languages (e.g., English, German, French, and Swedish). The TRQ has repeatedly been found to generate good psychometric properties.

Validated measures of depression and anxiety can be useful, both when planning and evaluating the treatment. While several questionnaires are available, including the 21-item Beck Depression Inventory (BDI; [Beck, Ward, Mendelson, Mock, & Erbaugh, 1961](#)) and the 40-item Spielberger State–Trait Anxiety Inventory (STAI; [Spielberger, Gorsuch, & Lushene,](#)

1970), great care must be taken to not exhaust or overburden patients with questionnaires. An efficient alternative is the Hospital Anxiety and Depression Scale (HADS; [Zigmond & Snaith, 1983](#)), which comprises only 14 items. Other advantages of the HADS are that it is more discrete and that it controls for complaints better explained by the somatic problem than does the BDI or STAI ([Herrmann, 1997](#)). It needs to be made clear that the HADS, and any other self-report inventory for that matter, is not used as a substitute for a proper psychiatric assessment. Psychiatric assessment is typically warranted in difficult cases.

Daily diaries can be used in research and in clinical practice to collect data on the characteristics of tinnitus. Tinnitus often fluctuates, sometimes even within a day, and occasionally, this aspect of the experience is not clear to the patient. Daily diaries need to be easily comprehended and unobtrusive while still providing usable data. One alternative is the Visual Analogue Scale (VAS), which is a collection of straight lines anchored at the ends with extremes (e.g., not at all bothersome, extremely bothersome) of the sensation, feeling, or response being measured ([Wevers & Lowe, 1990](#)). The strength of the method lies in the aggregation of several measurements (e.g., weekly average of tinnitus annoyance) and, hence, the approach has been recommended for use in research ([Axelsson, Coles, Erlandsson, Meikle, & Vernon, 1993](#)).

There are few publications on the use of structured interviews with tinnitus patients. Most clinicians follow an interview protocol, but few have evaluated if the questions posed are appropriate. The advantages of structured interviews, including detailed information, reliability, and predictive validity, have recently been pointed out empirically ([Andersson et al., 1999; Hiller, Goebel, & Schindelmann, 1999](#)).

Despite the common report of concentration difficulties among tinnitus sufferers, there have been few serious attempts to ascertain the utility of neuropsychological assessment. One exception is the work by McKenna ([McKenna et al., 1996; McKenna & Hallam, 1999](#)), who observed evidence of impaired cognitive processing in tinnitus patients. Clinically relevant information can be obtained from neuropsychological testing, but it is important to consider the eventual effects of hearing impairment on test performance. This field of research and practice needs to be further developed.

4. CBT for tinnitus

Among the earliest and most significant of contributions from psychologists to the understanding of tinnitus was [Hallam, Rachman, and Hinchcliffe's \(1984\)](#) psychological model. The authors suggested that the process of habituation characterizes the natural history of tinnitus. They conceived of patients' complaints of distress associated with tinnitus as a failure of habituation. Further, the model suggested that habituation to tinnitus follows the same rules as habituation to any other constant stimulus. They pointed out that habituation is slowed by factors such as a high level of tonic arousal and by the stimulus acquiring an emotive significance.

Recent theorizing based on animal experiments ([Jastreboff, Hazell, & Graham, 1994](#)) is in keeping with the assertions made by [Hallam et al. \(1984\)](#). However, models stemming from

these experiments tend to be neurophysiological in nature and thereby put less emphasis on the cognitive mechanisms involved in tinnitus perception. Yet, recent neurophysiological findings suggest that both attentional and emotional brain areas are associated with tinnitus activity (Mirz et al., 1999).

Taken together, these two models and their associated empirical findings suggest that CBT could be applied in the treatment of tinnitus and, in particular, to the distress that it causes. Relaxation techniques could be used to reduce arousal, and cognitive restructuring to overcome maladaptive cognitions and fears related to tinnitus. In addition, gradual exposure to feared situations (e.g., silence when tinnitus is highly salient or when confronting certain sounds) could help to foster habituation to tinnitus.

In this paper, the focus is on CBT for tinnitus (Sweetow, 1995). This is not, however, the only psychological treatment approach that has been applied. In fact, treatments that aim to reduce the annoyance caused by tinnitus all involve psychological and/or educational components. It is important to note that CBT for tinnitus does not represent one form of treatment, and that clinicians differ in what components they prefer. For example, some recommend relaxation techniques (Lindberg, Scott, Melin, & Lyttkens, 1988), whereas others rely mostly on cognitive therapy techniques (Wilson, Henry, & Nicholas, 1993). Overall, the application of CBT for tinnitus follows standard methods developed for other problems such as anxiety and pain (Hawton et al., 1989; Philips & Rachman, 1996). Hence, homework assignments between therapy sessions are scheduled, and a rationale is presented for each treatment component. In addition, the therapeutic relation between therapist and patient is collaborative in the sense that the outline of each session, and the treatment as a whole, is negotiated. Motivation to change habits and to alter behavior is crucial, and it is made clear to the patient that work is required for the treatment to have an effect. Table 1 presents an overview of a typical CBT treatment package.

This treatment package, depending on individual needs, is usually presented in 6–10 weekly sessions. Although individual treatment is often required, positive experiences of providing CBT in a small group setting have been reported (Kröner-Herwig et al., 1995). Despite evidence of no clear differences in efficacy between individual and group treatments when compared using meta-analysis (Andersson & Lyttkens, 1999), there has been no direct comparison of the two. Recently, an Internet-administrated self-help version of CBT was developed and tested in a controlled study (Andersson, Strömberg, et al., *In Press*). This approach was inspired by the beneficial effects found for self-help methods for other health-related problems such as insomnia (Mimeault & Morin, 1999), and also by a study in which we investigated the effects of Internet administrated self-help CBT for headache (Ström, Pettersson, & Andersson, 2000). Although the study by Andersson, Strömberg, et al. (*In Press*) had an initially high dropout rate, positive effects were found for those who completed the treatment and also in the 1-year follow-up.

On close reading, several of the published papers on psychological treatment for tinnitus share characteristics with the CBT treatment described above. For example, research on self-hypnosis (Attias et al., 1993) showed promising results and is compatible with the cognitive therapy component of CBT when imagery techniques are used. The effect of pure counseling without any other professional input has not been thoroughly investigated. However, as noted

Table 1

Summary of a CBT package for tinnitus

<i>Information</i> regarding tinnitus and a proper medical examination of potential causes and moderating factors. Test of hearing and measures of tinnitus loudness are also included.
<i>Functional analysis</i> of influencing factors on tinnitus annoyance. This includes medical, as well as psychosocial, factors.
<i>Advice regarding hearing loss</i> and remedy of this as far as possible. This includes not only referral for the fitting of hearing aid(s), but also behavioral advice in the form of “hearing tactics.” These are not only for the individual but also directed to the people close to the patient.
Use of environmental <i>sound enrichment</i> strategies for facilitation of habituation to tinnitus. This might include tapes or CDs, but more importantly advice and analysis of fluctuations in tinnitus and the risk associated with trying to mask (i.e., cover) the tinnitus.
<i>Applied relaxation</i> —a method by which the patient is gradually taught to quickly relax and to use self-control over bodily and mental sensations (e.g., stress). The aim is not to reduce tinnitus but to control the effects of tinnitus. The goal is to obtain a balanced state of mind. In association with the relaxation training imagery techniques are introduced (e.g., imagining a beach).
<i>Cognitive restructuring</i> of thoughts and beliefs associated with tinnitus. The patient is helped to identify the content of his thoughts and is learned ways to challenge or control those thoughts usually described as unhelpful, or even inaccurate. Attention diversion techniques can also be included, as well as imagery techniques. In the context of changing beliefs and thoughts, it is important to work towards acceptance of tinnitus and to foster the idea that tinnitus is not worth all the attention it gets. During later stages of the treatment, it can be valuable to work for reinterpretation of tinnitus into something more pleasant.
<i>Emotional reactions</i> are focused upon, especially fear and avoidance in relation to tinnitus. Not only can fear and avoidance lead to a negative view of tinnitus, but also strong emotional reactions can occasionally develop into panic-like attacks when the patient seeks to escape from tinnitus. Apart from advice regarding sound enrichment it can be important to deal with adverse reactions to silence (when this is a problem). On the other hand, some patients develop fear for everyday sounds (hyperacusis), and in these cases, it is important to gradually expose to environmental sounds.
<i>Problems with concentration</i> are often a source of great distress for the tinnitus sufferer and are targeted in the treatment. Although not well developed for tinnitus, methods for improving concentration can be used.
<i>Sleep hygiene</i> , bedtime and worry-time restriction, relaxation, and cognitive restructuring for patients with sleep problems (Morin, 1993). These methods are tailored according to the special needs of the tinnitus patient.
<i>Relapse prevention</i> advice regarding likely lapses in resistance towards the intrusive effects of tinnitus, as well as to prevent more permanent relapses. In brief, this includes a proper discussion of risk factors for developing more severe tinnitus and hearing loss and devising a plan for what to do should the tinnitus become worse. This can include returning to the relaxation training and to the sound enrichment strategies.

above, the effects of relaxation in isolation are minimal and work for few patients with tinnitus (Ireland, Wilson, Tonkin, & Platt-Hepworth, 1985). In practice, relaxation is often reported to be beneficial, but it has to be couched within an explanatory rationale, since a few patients may experience tinnitus as more bothering when they try to relax.

5. Empirical status of psychological treatments for tinnitus

While there has been intensive research activity in the pursuit of a cure for tinnitus, relatively little controlled research has been conducted on tinnitus management, including psychological treatment and CBT. However, in comparison, psychological treatment, and, in

particular, CBT, is among the most thoroughly validated treatment approaches to tinnitus, with at least 10 controlled studies. The effect of CBT for tinnitus has recently been reviewed elsewhere (Henry & Wilson, 2000); but, for the present purpose, a summary of the primary findings is appropriate.

Andersson and Lyttkens (1999) conducted the only published meta-analysis on psychological treatment of tinnitus. Outcomes of 18 studies, including a total of 24 samples and up to 700 subjects, were coded. Included were studies on cognitive/CBT treatment, relaxation, hypnosis, biofeedback, educational sessions, and problem solving. Effect sizes for perceived tinnitus loudness, annoyance, negative affect (e.g., depression), and sleep problems were calculated for randomized controlled studies, pre–post design studies, and follow-up results. Results showed strong to moderate effects on tinnitus annoyance for controlled studies ($d=0.86$), pre–post designs ($d=0.50$), and at follow-up ($d=0.48$). Results on tinnitus loudness were weaker and disappeared at follow-up. Lower effect sizes were also obtained for measures of negative affect and sleep problems. Exploratory analyses revealed that CBT was more effective than other psychological treatments on ratings of annoyance in the controlled studies. We concluded that psychological treatment for tinnitus is effective but that aspects such as depression and sleep problems may need to be targeted in future studies.

There is little evidence that CBT has any long-term effects on the perceived loudness of tinnitus. This is the case for most tinnitus treatments available. The effect of CBT on annoyance caused by tinnitus is clearer and has been replicated in many studies. Most importantly, the effect seems to be rather stable over time, for at least 1 year following treatment. In addition to the recent meta-analysis, which included an analysis of the long-term effects for an average follow-up duration of 5 months (yielding a decrease of annoyance), we have recently collected longitudinal data for an average period of 5 years posttreatment for a sample of tinnitus patients who had received CBT (Andersson, Vretblad, Larsen, & Lyttkens, 2001). Results showed decreases in annoyance and an increase in tolerance of tinnitus.

The effect of CBT on insomnia is well established in studies where insomnia has been the primary complaint (Morin, 1993). When similar methods have been used with tinnitus patients the result is less clear. Clinical experience attests to the usefulness of sleep management (McKenna, 2000), but we (Andersson & Lyttkens, 1999) were able to identify only two controlled studies in which sleep problems had been measured pre- and posttreatment (i.e., Jakes, Hallam, McKenna, & Hinchcliffe, 1992; Kröner-Herwig et al., 1995).

Since CBT is a validated treatment for anxiety and depression, its application to tinnitus is likely to have some beneficial effect on these negative emotions. This has been demonstrated, but since most tinnitus patients have subclinical levels of anxiety and depression, floor effects may have obscured treatment effects. However, in our meta-analysis, the effect size for measures of negative affect in the controlled studies was $d=0.41$ (Andersson & Lyttkens, 1999). Interestingly, treatment protocols differ in their emphasis on techniques derived from cognitive therapy for depression. Henry and Wilson (1998) conducted a study on the differential effects of cognitive techniques (attention control/imagery training vs. cognitive restructuring) and concluded that a combination of approaches was most beneficial. It is possible that negative affect is more influenced by cognitive therapy than by applied

relaxation. Unfortunately, lack of studies did not allow for this comparison to be made in our meta-analysis (Andersson & Lyttkens, 1999).

Tinnitus treatments often include different components, and this also pertains to CBT. This makes it difficult to ascertain which treatment ingredients are the most effective, which are necessary, and which can be omitted. Dismantling studies have yet to be conducted. As mentioned above though, the available evidence suggests that relaxation techniques alone are not beneficial (Ireland et al., 1985).

Given the problems associated with relying on only statistical significance when interpreting research findings (Sohn, 1996), it has become useful practice to calculate the proportion of clients showing a clinical significant improvement following treatment (Jacobson & Truax, 1991). In behavioral medicine research, this is often defined as a 50% decrease on the outcome measure of interest (Blanchard & Schwartz, 1988). In a recent study, Henry and Wilson (1999) reported that more than 70% of their patients reached this criterion on the TRQ. In most other studies, the proportion of significantly improved tinnitus patients has not been reported. However, it is unlikely to be as high as 70%. On average, an estimation of the effects of CBT in clinical practice is that about 30–50% improve significantly. It is desirable to develop and implement treatments in which the criteria of clinical significant improvement are met.

6. Final remarks

Because tinnitus patients often suffer from psychosocial difficulties and psychiatric comorbidity, the importance of psychological assessment in their management is clear. Whether this pertains to individuals with tinnitus in the general population is not known.

There is now evidence that CBT can be effective for alleviating the distress caused by tinnitus. More work needs to be done if we are to reach the level of success obtained for problems such as anxiety and depression. In addition, the relative merit of CBT over alternative psychological and medical treatment approaches is not known. Even less is known about the effects of separate treatment ingredients often combined in CBT (e.g., cognitive therapy vs. relaxation), and, as such, dismantling studies are needed. More attention also needs to be directed at the problems of insomnia and cognitive deficits associated with tinnitus. The latter may be unique in the sense that the possibility to mask tinnitus with background sound may interact with cognitive processing and attentional mechanisms, and knowledge of these relations may influence treatment. The efficacy of alternative, commonly practiced, psychological treatment approaches (such as psychodynamic therapy) needs to be established. Likewise, large-scale studies on the effects of CBT are needed since many of the published trials have had insufficient statistical power to detect any effects. In addition, outcome should be expressed in terms of clinical significant changes (Jacobson & Truax, 1991), as this is still not common practice. Little is known regarding sequencing effects and the possible dose–response relation between number of hours in treatment and outcome. In conclusion, it is clear that clinical psychology plays a role in the management of tinnitus and that continued efforts to establish multidisciplinary teams involving clinical psychologists is needed.

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References

- American Psychiatric Association (1994). *Diagnostic and statistical manual of mental disorders* (4th ed.). Washington, DC: American Psychiatric Press.
- Andersson, G. (1996a). The role of optimism in patients with tinnitus and in patients with hearing impairment. *Psychology and Health*, 11, 697–707.
- Andersson, G. (1996b). The benefits of optimism: a meta-analytic review of the life orientation test. *Personality and Individual Differences*, 21, 719–725.
- Andersson, G., Eriksson, J., Lundh, L.-G., & Lyttkens, L. (2000). Tinnitus and cognitive interference: a Stroop paradigm study. *Journal of Speech, Hearing, and Language Research*, 43, 1168–1173.
- Andersson, G., & Lyttkens, L. (1999). A meta-analytic review of psychological treatments for tinnitus. *British Journal of Audiology*, 33, 201–210.
- Andersson, G., Lyttkens, L., & Larsen, H. C. (1999). Distinguishing levels of tinnitus distress. *Clinical Otolaryngology*, 24, 404–410.
- Andersson, G., & McKenna, L. (1998). Tinnitus masking and depression. *Audiology*, 37, 174–182.
- Andersson, G., Strömberg, T., Ström, T., & Lyttkens, L. (In Press). Randomised controlled trial of Internet based cognitive behavior therapy for distress associated with tinnitus. *Psychosomatic Medicine*.
- Andersson, G., & Vretblad, P. (2000). Tinnitus and anxiety sensitivity. *Scandinavian Journal of Behaviour Therapy*, 29, 57–64.
- Andersson, G., Vretblad, P., Larsen, H.-C., & Lyttkens, L. (2001). Longitudinal follow-up of tinnitus complaints. *Archives of Otolaryngology—Head and Neck Surgery*, 127, 175–179.
- Attias, J., Shemsh, Z., Bleich, A., Solomon, Z., Bar-Or, G., Alster, J., & Sohmer, H. (1995). Psychological profile of help-seeking and non-help-seeking tinnitus patients. *Scandinavian Audiology*, 24, 13–18.
- Attias, J., Shemsh, Z., Sohmer, H., Gold, S., Shoam, C., & Faraggi, D. (1993). Comparison between self-hypnosis, masking and attentiveness for alleviation of chronic tinnitus. *Audiology*, 32, 302–312.
- Axelsson, A., Coles, R., Erlandsson, S., Meikle, M., & Vernon, J. (1993). Evaluation of tinnitus treatment: methodological aspects. *Journal of Audiological Medicine*, 2, 141–150.
- Beck, A. T., Ward, C. H., Mendelson, M., Mock, J., & Erbaugh, J. (1961). An inventory for measuring depression. *Archives of General Psychiatry*, 4, 561–571.
- Blanchard, E. B., & Schwartz, S. P. (1988). Clinically significant changes in behavioral medicine. *Behavioral Assessment*, 10, 171–188.
- Briner, W., Risey, J., Guth, P., & Noris, C. (1990). Use of the million clinical multiaxial inventory in evaluating patients with severe tinnitus. *American Journal of Otolaryngology*, 11, 334–337.
- Butcher, J. N., Dahlstrom, W. G., Graham, J. R., Tellegen, A., & Kaemmer, B. (1989). *MMPI-2: an administrative and interpretive guide*. Minneapolis, MN: University of Minnesota Press.
- Collet, L., Moussu, M. F., Disant, F., Ahami, T., & Morgon, A. (1990). Minnesota Multiphasic Personality Inventory in tinnitus disorders. *Audiology*, 29, 101–106.
- Collins, N.L., & Read, S. J. (1990). Adult attachment, working models, and relationship quality in dating couples. *Journal of Personality and Social Psychology*, 58, 644–663.
- Davies, A., & Rafie, E. A. (2000). Epidemiology of tinnitus. In R. S. Tyler (Ed.), *Tinnitus handbook* (pp. 1–23). San Diego: Singular. Thomson Learning.

- Dineen, R., Doyle, J., & Bench, J. (1997). Audiological and psychological characteristics of a group of tinnitus sufferers, prior to tinnitus management training. *British Journal of Audiology*, 31, 27–38.
- Dobie, R. A. (1999). A review of randomized clinical trials of tinnitus. *Laryngoscope*, 109, 1202–1211.
- Erlandsson, S. (2000). Psychological profile of tinnitus patients. In R. S. Tyler (Ed.), *Tinnitus handbook* (pp. 25–57). San Diego: Singular. Thomson Learning.
- Folmer, R. L., Greist, S. E., Meikle, M. B., & Martin, W. H. (1999). Tinnitus severity, loudness, and depression. *Otolaryngology, Head, and Neck Surgery*, 121, 48–51.
- Fowler, E. P. (1948). The emotional factor in tinnitus aurium. *Laryngoscope*, 58, 145–154.
- Gerber, K. E., Nehemkis, A. M., Charter, R. A., & Jones, H. C. (1985). Is tinnitus a psychological disorder? *International Journal of Psychiatry in Medicine*, 15, 81–87.
- Granqvist, P., Lantto, S., Ortiz, L., & Andersson, G. (2001). Adult attachment, perceived family support, and problems experienced by tinnitus patients. *Psychology and Health*, 16, 357–366.
- Halford, J. B. S., & Anderson, S. D. (1991). Anxiety and depression in tinnitus sufferers. *Journal of Psychosomatic Research*, 35, 383–390.
- Hallam, R. S., Rachman, S., & Hinchcliffe, R. (1984). Psychological aspects of tinnitus. In S. Rachman (Ed.), *Contributions to medical psychology* (pp. 31–53). Oxford: Pergamon.
- Hallberg, L.R.-M., & Erlandsson, S. I. (1993). Tinnitus characteristics in tinnitus complainers and noncomplainers. *British Journal of Audiology*, 27, 19–27.
- Hawton, K., Salkovskis, P. M., Kirk, J., & Clark, D. M. (1989). *Cognitive behaviour therapy for psychiatric problems*. Oxford: Oxford University Press.
- Heath, I. (1994). Tinnitus and health anxiety. *British Journal of Nursing*, 3, 502–505.
- Henry, J., & Wilson, P. (1999). Cognitive-behavioural therapy for tinnitus-related distress: an experimental evaluation of initial treatment and relapse prevention. In J. Hazell (Ed.), *Proceedings of the sixth international tinnitus seminar* (pp. 399–402). Cambridge, UK: Tinnitus and Hyperacusis Centre.
- Henry, J., & Wilson, P. (2000). Psychological management of tinnitus. In R. S. Tyler (Ed.), *Tinnitus handbook* (pp. 263–279). San Diego: Singular. Thomson Learning.
- Henry, J. L., & Wilson, P. H. (1998). An evaluation of two types of cognitive intervention in the management of chronic tinnitus. *Scandinavian Journal of Behaviour Therapy*, 27, 156–166.
- Herrmann, C. (1997). International experience with the Hospital Anxiety and Depression Scale—a review of validation data and clinical results. *Journal of Psychosomatic Research*, 42, 17–41.
- Hiller, W., & Goebel, G. (1992). Psychiatric disorders and their degree of severity in patients with severe tinnitus and pain syndromes. In J.-M. Aran & R. Dauman (Eds.), *Tinnitus 91. Proceedings of the fourth international tinnitus seminar* (pp. 441–444). Amsterdam: Kugler Publications.
- Hiller, W., Goebel, G., & Schindelmann, U. (1999). Developing a structured interview to assess audiological, aetiological and psychological variables of tinnitus. In J. Hazell (Ed.), *Proceedings of the sixth international tinnitus seminar* (pp. 277–282). Cambridge, UK: Tinnitus and Hyperacusis Centre.
- Hiller, W., Janca, A., & Burke, K. C. (1997). Association between tinnitus and somatoform disorders. *Journal of Psychosomatic Research*, 43, 613–624.
- Ireland, C. E., Wilson, P. H., Tonkin, J. P., & Platt-Hepworth, S. (1985). An evaluation of relaxation training in the treatment of tinnitus. *Behaviour Research and Therapy*, 23, 423–430.
- Jacobson, N. S., & Truax, P. (1991). Clinical significance: a statistical approach to defining meaningful change in psychotherapy research. *Journal of Consulting and Clinical Psychology*, 59, 12–19.
- Jakes, S. C., Hallam, R. S., McKenna, L., & Hinchcliffe, R. (1992). Group therapy for medical patients: an application to tinnitus. *Cognitive Therapy and Research*, 16, 67–82.
- Jastreboff, P. J., Hazell, J. W. P., & Graham, R. L. (1994). Neurophysiological model of tinnitus: dependence of the minimal masking level on treatment outcome. *Hearing Research*, 80, 216–232.
- Johnston, M., & Walker, M. (1996). Suicide in the elderly. Recognizing the signs. *General Hospital Psychiatry*, 18, 257–260.
- Kerns, R. D., Turk, D. C., & Rudy, T. E. (1985). The West Haven–Yale Multidimensional Pain Inventory (WHYMPI). *Pain*, 23, 345–356.

- Kirsch, C. A., Blanchard, E. B., & Parnes, S. M. (1989). Psychological characteristics of individuals high and low in their ability to cope with tinnitus. *Psychosomatic Medicine*, 51, 209–217.
- Kröner-Herwig, B., Hebing, G., Van Rijn-Kalkmann, U., Frenzel, A., Schilkowsky, G., & Esser, G. (1995). The management of chronic tinnitus. Comparison of a cognitive-behavioural group training with yoga. *Journal of Psychosomatic Research*, 39, 153–165.
- Lewis, J., Stephens, D., & Huws, D. (1992). Suicide in tinnitus. *Journal of Audiological Medicine*, 1, 30–37.
- Lewis, J. E., & Stephens, S. D. G. (1996). Parasuicide and tinnitus. *Journal of Audiological Medicine*, 4, 34–43.
- Lewis, J. E., Stephens, S. D. G., & McKenna, L. (1994). Tinnitus and suicide. *Clinical Otolaryngology*, 19, 50–54.
- Lindberg, P., Scott, B., Melin, L., & Lyttkens, L. (1988). Behavioural therapy in the clinical management of tinnitus. *British Journal of Audiology*, 22, 265–272.
- McFadden, D. (1982). *Tinnitus. Facts, theories, and treatments*. Washington, DC: National Academy Press.
- McKenna, L. (2000). Tinnitus and insomnia. In R. S. Tyler (Ed.), *Tinnitus handbook* (pp. 59–84). San Diego: Singular. Thomson Learning.
- McKenna, L., & Andersson, G. (1998). Hearing disorders. In M. Hersen & A. Bellack (Eds.), *Comprehensive clinical psychology* (pp. 69–83). Oxford: Pergamon.
- McKenna, L., & Hallam, R. (1999). A neuropsychological study of concentration problems in tinnitus patients. In J. Hazell (Ed.), *Proceedings of the sixth international tinnitus seminar* (pp. 108–113). Cambridge, UK: Tinnitus and Hyperacusis Centre.
- McKenna, L., Hallam, R. S., & Hinchcliffe, R. (1991). The prevalence of psychological disturbance in neurotology outpatients. *Clinical Otolaryngology*, 16, 452–456.
- McKenna, L., Hallam, R. S., & Shurlock, L. (1996). Cognitive functioning in tinnitus patients. In G. E. Reich & J. A. Vernon (Eds.), *Proceedings of the fifth international tinnitus seminar 1995* (pp. 589–595). Portland: American Tinnitus Association.
- Meikle, M. B., & Greist, S. E. (1989). Gender-based differences in characteristics of tinnitus. *Hearing Journal*, 42, 68–76.
- Mimeault, V., & Morin, C. M. (1999). Self-help treatment for insomnia: bibliotherapy with and without professional guidance. *Journal of Consulting and Clinical Psychology*, 67, 511–519.
- Mirz, F., Pedersen, C. B., Ishizu, K., Johannsen, P., Ovesen, T., Stødkilde-Jørgensen, H., & Gjedde, A. (1999). Positron emission tomography of cortical centres of tinnitus. *Hearing Research*, 134, 133–144.
- Morin, C. M. (1993). *Insomnia. Psychological assessment and management*. New York: Guilford Press.
- Newman, C. W., Wharton, J. A., & Jacobson, G. P. (1997). Self-focused and somatic attention in patients with tinnitus. *Journal of the American Academy of Audiology*, 8, 143–149.
- Noble, W. (2000). Self-reports about tinnitus and about cochlear implants. *Ear and Hearing*, 21, 50S–59S.
- Philips, H. C., & Rachman, S. (1996). *The psychological management of pain*. New York: Springer.
- Reiss, S., Peterson, R. A., Gurksy, D. M., & McNally, R. J. (1986). Anxiety sensitivity, anxiety frequency and the prediction of fearfulness. *Behaviour Research and Therapy*, 34, 1–8.
- Russo, J. E., Katon, W. J., Sullivan, M. D., Clark, M. R., & Buchwald, D. (1994). Severity of somatization and its relationship to psychiatric disorders and personality. *Psychosomatics*, 35, 546–556.
- Sanchez, L., & Stephens, D. (1997). A tinnitus problem questionnaire. *Ear and Hearing*, 18, 210–217.
- Scheier, M. F., & Carver, C. S. (1985). Optimism, coping, and health: assessment and implications of generalized outcome expectancies. *Health Psychology*, 4, 219–247.
- Schulman, A. (1995). A final common pathway for tinnitus—the medial temporal lobe system. *International Tinnitus Journal*, 1, 115–126.
- Simpson, R. B., Nedzelski, J. M., Barber, H. O., & Thomas, M. R. (1988). Psychiatric diagnoses in patients with psychogenic dizziness or severe tinnitus. *Journal of Otolaryngology*, 17, 325–330.
- Sohn, D. (1996). Publication bias in the evaluation of psychotherapy efficacy in reviews of the research literature. *Clinical Psychology Review*, 16, 147–156.
- Spielberger, C. D., Gorsuch, R. L., & Lushene, R. E. (1970). *Test manual for the State-Trait Anxiety Inventory*. Palo Alto, CA: Consulting Psychologists Press.

- Stephens, D., Erlandsson, S., Sanchez, L., & Huws, D. (1992). Some psychological aspects of tinnitus. In J.-M. Aran, & R. Dauman (Eds.), *Tinnitus 91. Proceedings of the fourth international tinnitus seminar* (pp. 433–439). Amsterdam: Kugler Publications.
- Stephens, S. D. G., & Hallam, R. S. (1985). The Crown–Crisp experiential index in patients complaining of tinnitus. *British Journal of Audiology*, 19, 151–158.
- Stouffer, J. L., & Tyler, R. S. (1992). Ratings of psychological changes pre- and post-tinnitus onset. In J.-M. Aran & R. Dauman (Eds.), *Tinnitus 91. Proceedings of the fourth international tinnitus seminar* (pp. 449–452). Amsterdam: Kugler Publications.
- Ström, L., Pettersson, R., & Andersson, G. (2000). A controlled trial of self-help treatment of recurrent headache conducted via the internet. *Journal of Consulting and Clinical Psychology*, 68, 722–727.
- Sullivan, M., Katon, W., Russo, J., Dobie, R., & Sakai, C. (1994). Coping and marital support as correlates of tinnitus disability. *General Hospital Psychiatry*, 16, 259–266.
- Sullivan, M. D., Katon, W., Dobie, R., Sakai, C., Russo, J., & Harrop-Griffiths, J. (1988). Disabling tinnitus. Association with affective disorders. *General Hospital Psychiatry*, 10, 285–291.
- Sweetow, R. W. (1995). The evolution of cognitive–behavioral therapy as an approach to tinnitus management. *International Tinnitus Journal*, 1, 61–65.
- Taylor, S., Koch, W. J., McNally, R. J., & Crockett, D. J. (1992). Conceptualizations of anxiety sensitivity. *Psychological Assessment*, 4, 2245–2250.
- Tyler, R. S. (1993). Tinnitus disability and handicap questionnaires. *Seminars in Hearing*, 14, 377–383.
- Tyler, R. S. (Ed.) (2000). *Tinnitus handbook*. San Diego: Singular. Thomson Learning.
- Tyler, R. S., & Babin, R. W. (1993). Tinnitus. In C. W. Cummings, J. M. Fredrickson, L. A. Harker, C. J. Krause, & D. E. Schuller (Eds.), *Otolaryngology—head and neck surgery* (pp. 3031–3053). St. Louis: CV Mosby.
- Tyler, R. S., & Baker, L. J. (1983). Difficulties experienced by tinnitus sufferers. *Journal of Speech and Hearing Disorders*, 48, 150–154.
- Wilson, P. H., Henry, J., Bowen, M., & Haralambous, G. (1991). Tinnitus Reaction Questionnaire: psychometric properties of a measure of distress associated with tinnitus. *Journal of Speech and Hearing Research*, 34, 197–201.
- Wilson, P. H., Henry, J. L., & Nicholas, M. K. (1993). Cognitive methods in the management of chronic pain and tinnitus. *Australian Psychologist*, 28, 172–180.
- Wevers, M. E., & Lowe, N. K. (1990). A critical review of visual analogue scales in the measurement of clinical phenomena. *Research in Nursing and Health*, 13, 227–236.
- Zigmond, A. S., & Snaith, R. P. (1983). The Hospital Anxiety and Depression Scale. *Acta Psychiatrica Scandinavica*, 67, 361–370.
- Zöger, S., Holgers, K. M., & Svedlund, J. (2001). Psychiatric disorders in tinnitus patients without severe hearing impairment: 24 month follow-up of patients at an audiological clinic. *Audiology*, 40, 133–140.